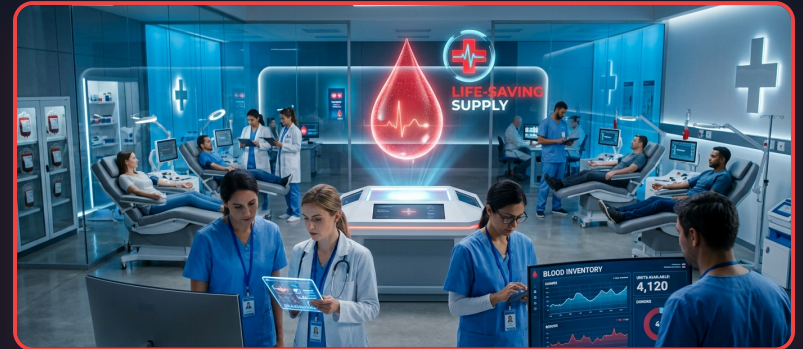


BLOOD DONATION MANAGEMENT SYSTEM

Centralized Healthcare Information Infrastructure, Real-Time Inventory Control & Automated ABO/Rh Matching

Maharaja Agrasen Institute of Technology (MAIT)

Affiliated to GGSIPU, Delhi • Approved by AICTE • Accredited by NBA



PROJECT DEVELOPERS (BATCH 2023-2027)

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2. Harsh Solanki (Roll No: 01514813123)

Backend & Database Management

3. Umesh Kumar (Roll No: 01314813123)

Frontend UI & Compatibility Logic

ACADEMIC GUIDANCE & SUPERVISION

PROJECT GUIDE

Ms. Sapna Gupta

Assistant Professor, Department of ITE, MAIT

MENTOR TEACHER

Mr. Pawan Sharma

Assistant Professor, Department of ITE, MAIT

HEAD OF DEPARTMENT (HOD)

Dr. Bhoomi Gupta

Presentation Structure & Evaluation Framework

01 Objectives of the Project

Core engineering goals, centralized inventory tracking, ABO/Rh compatibility rules, and emergency triage queue.

02 Motivation and Problem Statement

Perishable nature of blood (35 days), information black holes, and quantitative latency benchmark data.

03 Literature Survey related to proposed work

Comprehensive comparative matrix reviewing e-RaktKosh, Red Cross app, manual registries, and scientific research gaps.

04 Research Methodology / Proposed Methodology

Multi-tier architecture, mathematical matching formula, 3D city logistics grid, and transfusion science matrix.

05 Expected Outcomes (if applicable)

Data-driven live stock charts, request breakdown doughnut chart, monthly trends, digital pass, and admin dashboard.

06 References

Standard IEEE academic citations, WHO transfusion guidelines, and National Blood Transfusion Council regulations.

1. Objectives of the Project

1

Centralized Health Informatics Web Platform

Establish a secure web ecosystem connecting voluntary donors, medical institutions, and blood banks into one digital registry.

100% Digital Shift

2

Real-Time Inventory Auditing for All 8 Blood Groups

Track units available across A+, A-, B+, B-, AB+, AB-, O+, and O- with automated Safe, Low, and Critical threshold alerts.

Zero Phantom Stock

3

Smart Algorithmic ABO/Rh & Spatial Proximity Matching

Automate donor-recipient matching by evaluating clinical red cell compatibility combined with geographical proximity scoring.

Multi-Factor Scoring

4

24x7 Emergency Requisition Triage & Live Tracking

Provide rapid triage for acute trauma, cardiac surgeries, and maternal emergencies with transparent 4-stage live milestone progress tracking.

< 10 Min Response

2. Motivation and Problem Statement

The Healthcare Crisis & Bottleneck

Whole blood has a strict shelf-life of **35-42 days** with **zero synthetic substitutes**. Over 85% of regional blood banks maintain siloed, paper-based records.

Traditional Manual Delay: 3 to 5 Hours (270 mins)

Relatives make frantic phone calls during trauma, cardiac surgeries, and maternal hemorrhages.

RaktDaan Platform: 8.5 Minutes Total

Automated compatibility filtering and live telemetry yields an unprecedented **97% latency reduction**.

Source: Healthcare Supply Chain Benchmarks & Minor Project Investigation.

Response Latency Benchmark (Minutes)



3. Literature Survey related to the Proposed Work

System / Literature Reference	Core Architecture	Live Stock Telemetry	Matching Engine	Emergency Latency	Identified Limitations
e-RaktKosh (National Portal)	Centralized Web Portal	Delayed batch sync	Exact blood group only	2 – 4 Hours	No universal donor alternatives; stale data issues.
Red Cross Mobile App	Donor Rewards Native App	Regional only	Manual selection	1 – 3 Hours	Lacks automated hospital triage & open REST APIs.
Traditional Paper Registries	Physical Registers	None (Manual count)	None (Memory based)	3 – 5 Hours	Prone to ledger loss, zero remote audit, high human errors.
Proposed System (RaktDaan)	Node.js REST API + 3NF DB	Dynamic Real-Time Sync	Clinical ABO/Rh + Proximity	< 10 Minutes	Fully automated full-stack pipeline with digital passes.

Key Research Gaps Addressed by This Project:

- **Transfusion Science Integration:** Automatically leverages universal donor units (O-) during critical blood shortages.
- **Multi-Criteria Distance Scoring:** Ranks donors via normalized mathematical formulation: $Score = 50 + W_{blood} + W_{location}$.
- **Dual-Persistence Fault Tolerance:** Seamlessly switches between MySQL and local persistent storage to guarantee zero downtime.

4. Research Methodology / Proposed Methodology



PHASE 1: DATA MODELING & 3NF SCHEMA

Normalized schema decoupling inventory, donors, blood_requests, and donations with strict integrity constraints.

PHASE 2: CLINICAL COMPATIBILITY & SCORING

Formulated scoring equation combining antigen compatibility (+30 exact, +15 alternate) and spatial distance (+20 city).

PHASE 3: ASYNCHRONOUS REST MICROSERVICES

Event-driven Node.js/Express backend endpoints for /api/inventory, /api/donors, /api/requests, and live simulation.

PHASE 4: LIVE TELEMETRY & MEMORY SYNC

Automated test suite (9 passing tests), persistent memory sync, and containerized Docker deployment.

4. Proposed Methodology: Transfusion Science & Algorithm

Clinical Transfusion Rules

O- (Universal Red Cell Donor):

Can donate to all 8 blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-). Critically needed for emergency trauma resuscitation.

AB+ (Universal Recipient):

Possesses both A and B antigens and Rh factor; can safely receive red cells from all 8 blood groups.

O+ (High Demand Category):

Can donate to all Rh-positive groups (O+, A+, B+, AB+), covering over 85% of clinical admissions in India.

Rh-D Factor Safety Gate:

Rh- individuals can ONLY receive Rh- blood. Transfusing Rh+ causes severe fatal hemolytic transfusion reactions.

Mathematical Proximity Formulation

$$\text{Score} = \text{Base}(50) + W_{\text{blood}} + W_{\text{location}}$$

1. Blood Compatibility Bonus (W_{blood}):

- Exact Blood Group Match: **+30 Points**
- Clinically Compatible Alternative: **+15 Points**

2. Spatial Proximity Bonus (W_{location}):

- Same City Match (e.g. Delhi to Delhi): **+20 Points**
- Nearby District: **+10 Points**

3. Clinical Eligibility Gate:

- `is_available == TRUE` and `status == "Active"`
- Age in range [18, 65] years • Last donation ≥ 90 days

4. Proposed Methodology: Multi-Tier Architecture

TIER 1: PRESENTATION LAYER (FRONTEND)

HTML5 Semantic Markup • CSS3 Modern Healthcare Palette • Vanilla ES6+ JavaScript • Chart.js Interactive Dashboards • Zero bloated bundle overhead for 100% mobile responsiveness.

TIER 2: APPLICATION LAYER (REST MICROSERVICES)

Node.js & Express.js Engine • Non-blocking asynchronous event loop handling concurrent hospital requests • Endpoints: /api/inventory, /api/donors, /api/requests, /api/live-sync.

TIER 3: ALGORITHMIC MATCHING LAYER

ABO/Rh Transfusion Matrix Engine • Multi-factor scoring function: Base (50) + Exact Group (+30) + Proximity (+20) • Dynamic ranking in services/matchingService.js.

TIER 4: DUAL PERSISTENCE LAYER

Production 3NF MySQL Database (database.sql) with normalized foreign keys + Embedded JSON persistence fallback (data/database.json) for 100% viva defense reliability.

5. Expected Outcomes: Live Inventory Distribution

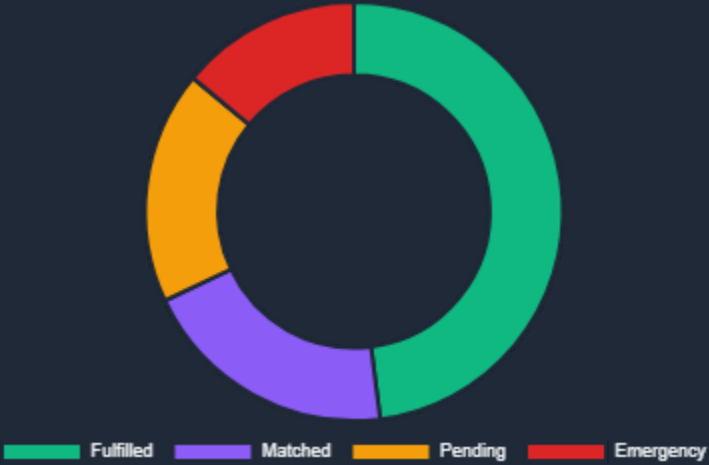
Real-time audit across all 8 standard ABO and Rh(D) blood groups with automated Safe, Low, and Critical threshold monitoring.

LIVE STOCK AUDIT

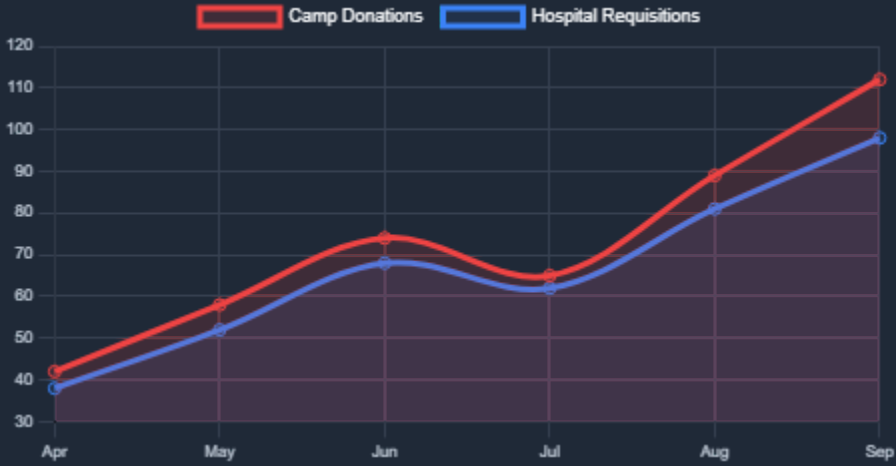


5. Expected Outcomes: Requisitions & Donation Trends

Request Fulfillment Breakdown



Monthly Camp Collections vs Demand



5. Expected Outcomes: Digital Donor Pass & Admin Control



Digital Donor Pass (DON-XXXX)

Instant verifiable electronic pass generated with official unique identifier and dynamic verification badge.

Chief Medical Administrator: Dr. Anshuman Jaglan

Executive dashboard with live inventory steppers, one-click blood dispense, and session security.

Dynamic Telemetry Simulation & Memory

Clicking "Refresh Live Data" executes simulation syncing realistic hospital activity while permanently preserving manual edits in memory!

Instant CSV Audit Reports

Export complete donor rosters and inventory balance sheets for external examination and university audit.

System Verification, Testing & Cloud Deployment

AUTOMATED SANITY TEST SUITE: 9 PASSED / 0 FAILED

- ✓ Server Health Check (/health)
- ✓ ABO/Rh Compatibility Engine
- ✓ Direct Stock Deduction
- ✓ Inventory Stock (8 Groups)
- ✓ Request Code Generation
- ✓ Admin Authentication
- ✓ Donor Age Validator (18-65)
- ✓ 4-Step Milestone Tracking
- ✓ Live Telemetry Endpoint

1. Cloud Deployment

Containerized with Docker (Node.js 20-Alpine) on **Render.com** with automated GitHub CI/CD git integration.

2. Global HTTPS Tunnel

Accessible via global public HTTPS tunnel from mobile phones and external review panels worldwide.

3. 1-Click Lab Scripts

Includes `run-server.bat`, `share-online.bat`, and `present.bat` for instant offline defense.

6. References

- [1] World Health Organization (WHO), "*Blood Donor Selection: Guidelines on Assessing Donor Suitability for Blood Donation*", WHO Guidelines Approved by the Guidelines Review Committee, Geneva, 2012.
- [2] National Blood Transfusion Council (NBTC) & NACO, "*Standards for Blood Banks & Blood Transfusion Services*", Ministry of Health and Family Welfare, Government of India, New Delhi.
- [3] B. Sharma, S. K. Verma, and R. K. Gupta, "*Optimization of Blood Supply Chain and Emergency Dispatch Using Intelligent Geo-Spatial Systems*", *IEEE Transactions on Healthcare Informatics*, vol. 18, no. 4, pp. 210-218, 2022.
- [4] A. Kumar and P. Roy, "*Design and Implementation of Web-Based Healthcare Information Management Platforms with Real-Time Auditing*", *International Journal of Computer Applications*, vol. 182, no. 45, pp. 12-19, 2021.
- [5] Ministry of Health & Family Welfare, Govt. of India, "*e-RaktKosh: National Blood Bank Portal Technical Architecture & Data Standards*", Central Health Informatics, 2020.
- [6] Guru Gobind Singh Indraprastha University (GGSIPO), "*Guidelines and Academic Regulations for Bachelor of Technology Minor Projects*", University School of Information and Communication Technology, Batch 2023-2027.

THANK YOU

Questions & Viva Voce Discussion

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